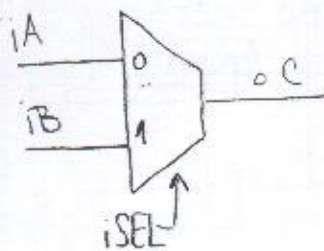
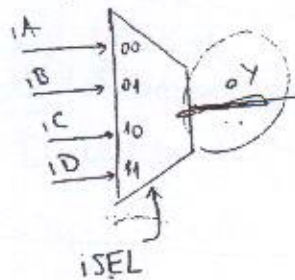


MUX (MULTIPLIKACEP)

$oC \leq iA$ when $iSEL = '0'$ else iB ;



$oY \leq iA$ when $iSEL = "00"$ else
 iB when $iSEL = "01"$ else
 iC when $iSEL = "10"$ else
 iD ;



POMERAČ $\begin{cases} \text{LOGIČKO} \\ \text{ARITMETIČKO (ČUVA ZNAK!) } \end{cases}$



1MERI: LOGIČKO DESNO ZA 1:

$iA: \underline{3} \underline{2} \underline{1} \underline{0}$

$iB: \emptyset \underline{3} \underline{2} \underline{1}$

$iB \leq '0' \& iA(3 \text{ downto } 1)$

LOGIČKO DESNO ZA 2:

$iA: \underline{3} \underline{2} \underline{1} \underline{0}$

$iB: \emptyset \emptyset \underline{3} \underline{2}$

$iB \leq "00" \& iA(3 \text{ downto } 2)$

LOGIČKO LEVO ZA 1:

$iA: \underline{3} \underline{2} \underline{1} \underline{0}$

$iB: \underline{2} \underline{1} \underline{0} \emptyset$

$iB \leq iA(2 \text{ downto } 0) \& '0'$

LOGIČKO LEVO ZA 2:

$iA: \underline{3} \underline{2} \underline{1} \underline{0}$

$iB: \underline{1} \underline{0} \emptyset \emptyset$

$iB \leq iA(1 \text{ downto } 0) \& "00"$

$iA: \underline{3} \underline{2} \underline{1} \underline{0}$
 $iB: \emptyset \emptyset \emptyset \underline{3} \underline{2} \underline{1} \underline{0} \emptyset$

ARITMETIČKI DESNO ZA 1:

$iA: \underline{3} \underline{2} \underline{1} \underline{0}$

$iB: \underline{3} \underline{3} \underline{2} \underline{1}$

$iB \leq iA(3) \& iA(3 \text{ downto } 1)$

ARITMETIČKI DESNO ZA 2:

$iA: \underline{3} \underline{2} \underline{1} \underline{0}$

$iB: \underline{3} \underline{3} \underline{3} \underline{2}$

$iB \leq iA(3) \& iA(3) \& iA(3 \text{ downto } 2)$

ARITMETIČKI LEVO ZA 1:

$iA: \underline{3} \underline{2} \underline{1} \underline{0}$

$iB: \underline{2} \underline{1} \underline{0} \emptyset$

$iB \leq iA(2 \text{ downto } 0) \& '0'$

PAZNJA:

iA ima 4 bita $\rightarrow iA(3 \text{ downto } 0)$
 iA ima 8 bita $\rightarrow iA(7 \text{ downto } 0)$

$iA: \underline{\hspace{1cm}} \underline{\hspace{1cm}} \underline{\hspace{1cm}} \underline{\hspace{1cm}} \underline{\hspace{1cm}} \underline{\hspace{1cm}} \underline{\hspace{1cm}} \underline{\hspace{1cm}}$
 $iA(5 \text{ downto } 2)$

0ZERO < '1' when iA = "0000" else '0';



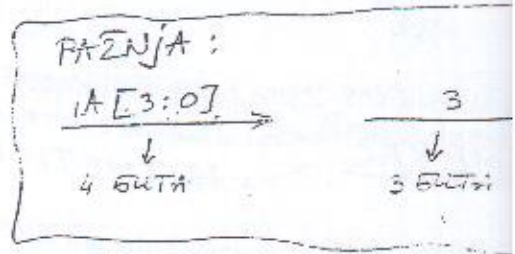
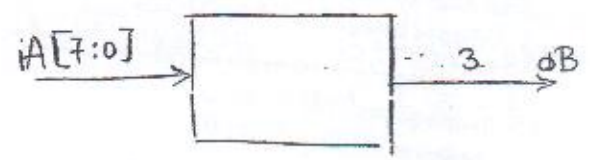
SABIRACI

iA -----
iB -----
oC <= iA + iB

iA : -----
iB : -----
oC <= iA + ('0' & iB)

iA -----
iB -----
oC <= iA + ("0000" & iB)

PRIORITETNI KODER



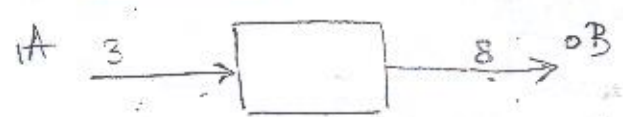
oB <= "111" when iA(7) = '1' else
 "110" when iA(6) = '1' else
 "101" when iA(5) = '1' else
 "100" when iA(4) = '1' else
 "011" when iA(3) = '1' else
 "010" when iA(2) = '1' else
 "001" when iA(1) = '1' else
 "000"

TRAŽIMO KOJA JE "NAJLEVIJA" JEDINICA (PRIORITETNA) i U IZLAZ UPİŠEMO BROJ MESTA NA KOM JE TA JEDINICA (U TROBITNOJ PREDSTAVI)

PR: 00101011 → "NAJLEVIJA" 7 6 5 4 3 2 1 0

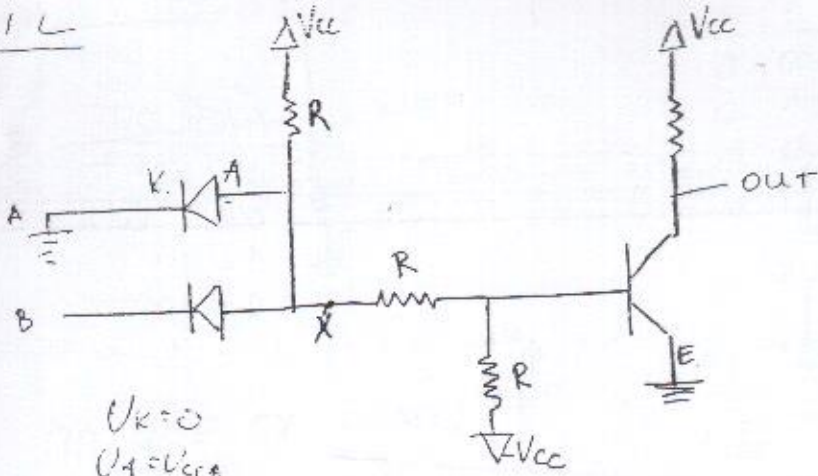
JEDINICA JE NA 5. MESTU, ZNAČI U IZLAZ IDE BROJ 5 U TROBITNOJ PREDSTAVI (101)

DEKODER



oB <= "10000000" when iA = "111" else
 "01000000" when iA = "110" else
 "00100000" when iA = "101"

U1L



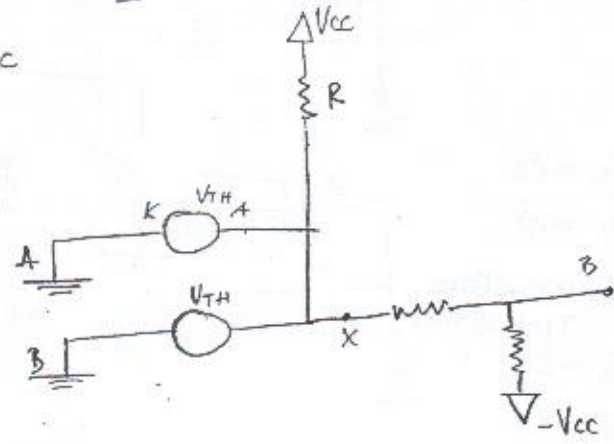
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

$U_K = 0$
 $U_A = V_{cc}$

00 $U_{AK} = V_{cc} > V_{TH}$
 D_1 ускб, D_2 ускб

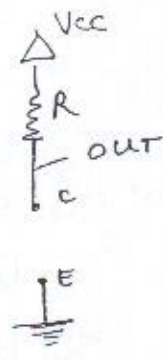
D_1 : $U_A = V_{cc}$
 $U_K = 0$
 $U_{AK} = V_{cc} > V_{TH}$
 D_1 проводи

D_2 : $U_A = V_{cc}$
 $U_K = 0$
 $U_{AK} = V_{cc} > V_{TH}$
 D_2 проводи



$V_X = V_{TH}$
 $U_B < 0$
 $U_E = 0$
 $U_{BE} < 0$

Т ускб у чен

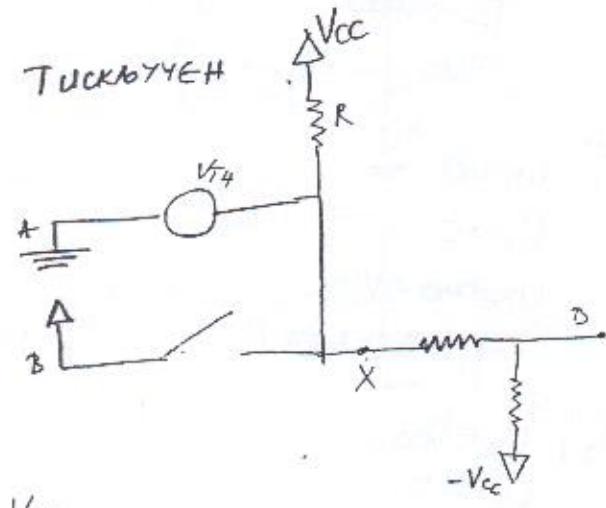


$OUT = V_{cc} = '1'$

01 D_1 ускб, D_2 укб.

D_1 : $U_A = V_{cc}$
 $U_K = 0$
 $U_{AK} = V_{cc} > V_{TH}$
 проводи

D_2 : $U_A = V_{cc}$
 $U_K = V_{cc}$
 $U_{AK} = 0 < V_{TH}$
 не проводи



$V_X = V_{TH}$

$U_B < 0$
 $U_E = 0$
 $U_{BE} < 0$

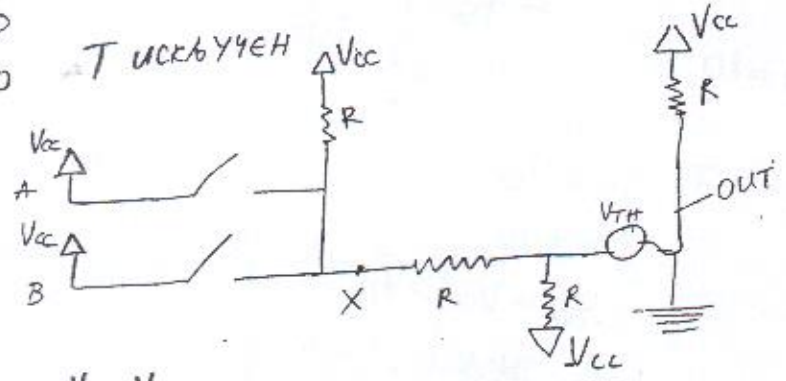
Т ускб у чен

$OUT = V_{cc} = '1'$

01 = 10

11 D_1 укб, D_2 укб.

D_1 : $U_A = V_{cc}$
 $U_K = V_{cc}$
 $U_{AK} = 0 < V_{TH} \Rightarrow$ не проводи



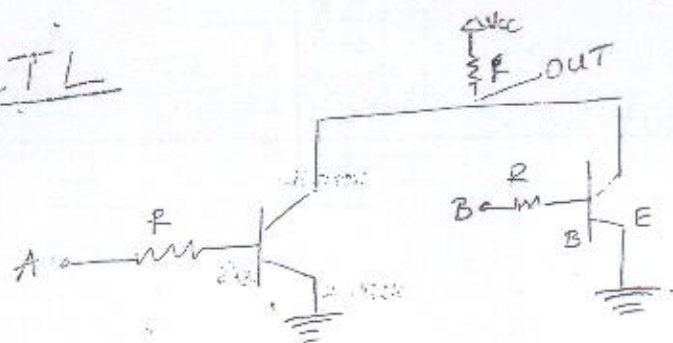
$V_X = V_{cc}$

$U_{BE} \geq V_{TH}$

Т не укрб

$OUT = '0'$

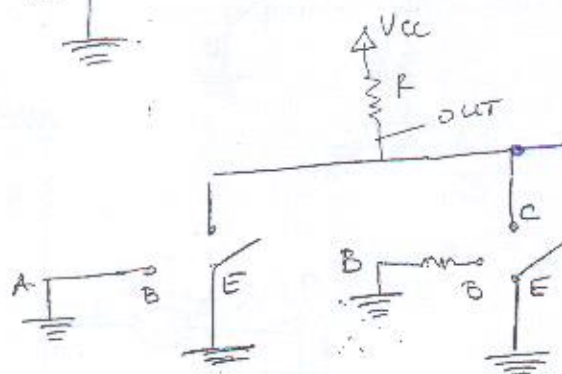
RTL



A	B	OUT
0	0	1
0	1	0
1	0	0
1	1	0

00 $T_1: U_B = 0$
 $U_E = 0$
 $U_{BE} = 0 < U_{TH}$
 $T_1 \rightarrow U_{CKB}$

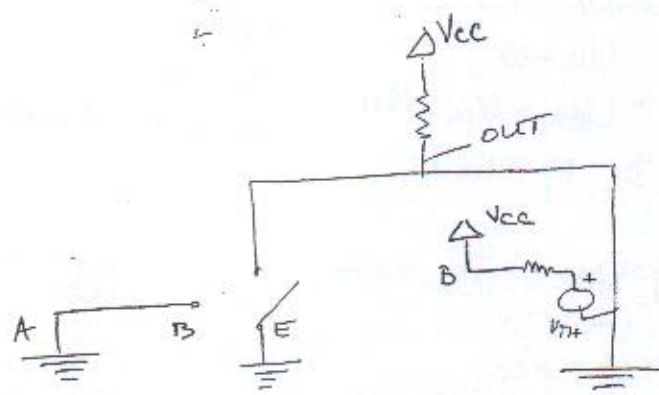
$T_2: U_B = 0$
 $U_E = 0$
 $U_{BE} = 0 < U_{TH}$
 $T_2 \rightarrow U_{CKB}$



OUT = Vcc = 1

01 $T_1: U_B = 0$
 $U_E = 0$
 $U_{BE} = 0 < U_{TH}$
 $T_1 \rightarrow U_{CKB}$

$T_2: U_B = V_{cc}$
 $U_E = 0$
 $U_{BE} = V_{cc} > U_{TH}$
 $T_2 \rightarrow U_{CKB}$

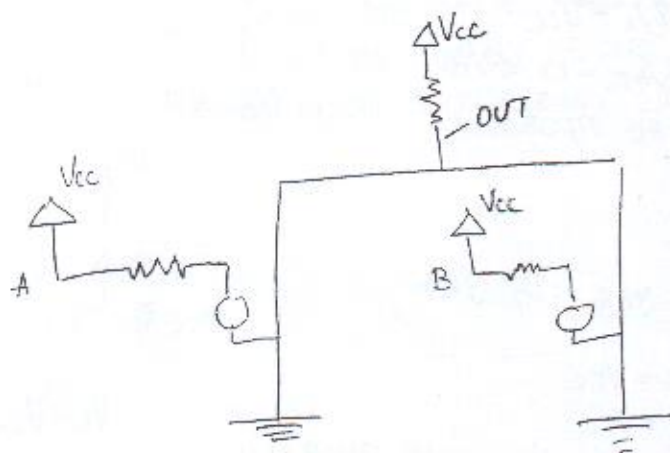


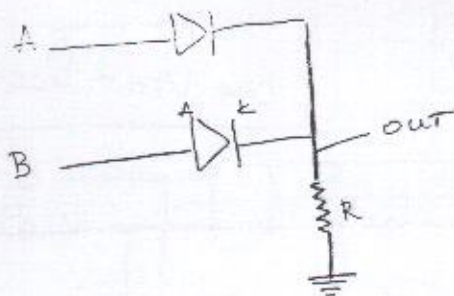
OUT = "0"

01=10

11 $T_1: U_B = V_{cc}$
 $U_E = 0$
 $U_{BE} = V_{cc} > U_{TH}$
 $T_1 \rightarrow U_{CKB}$

$T_2: U_B = V_{cc}$
 $U_E = 0$





A	B	OUT
0	0	0
0	1	1
1	0	1
1	1	1

00 ПП: ОБЕ СУ ИСКЛЮЧЕНЕ

$$D_1: U_A = 0$$

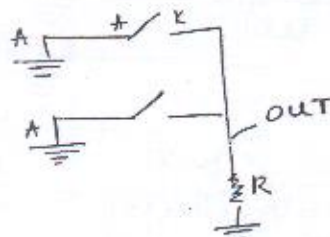
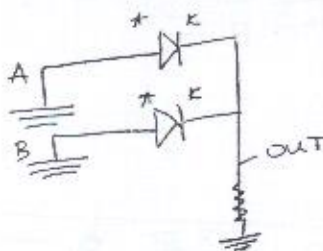
$$U_K = 0$$

$U_{AK} = 0 < V_{TH}$ не проводи

$$D_2: U_A = 0$$

$$U_K = 0$$

$U_{AK} = 0 < V_{TH}$ не проводи



OUT = "0"

01 ПП: D_1 ИСКЛЮЧЕНА, D_2 УКЛЮЧЕНА

$$D_1: U_A = 0$$

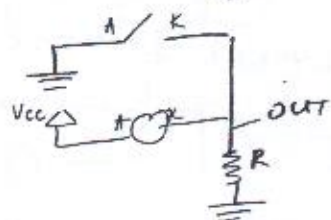
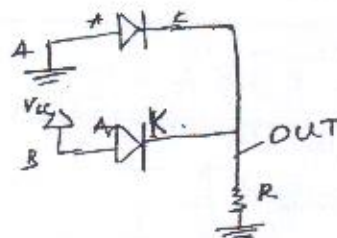
$$U_K = 0$$

$U_{AK} = 0 < V_{TH}$ не проводи

$$D_2: U_A = V_{CC}$$

$$U_K = 0$$

$U_{AK} = V_{CC} > V_{TH}$ проводи



OUT = "1"

01 = 10

ПП: D_1 УКЛЮЧЕНА, D_2 - УКЛЮЧЕНА

$$D_1: U_A = V_{CC}$$

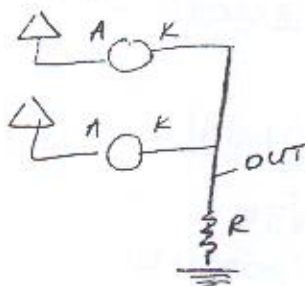
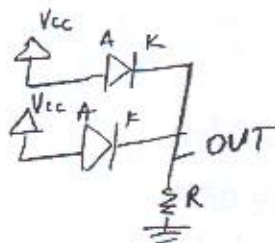
$$U_K = 0$$

$U_{AK} = V_{CC} > V_{TH}$ проводи

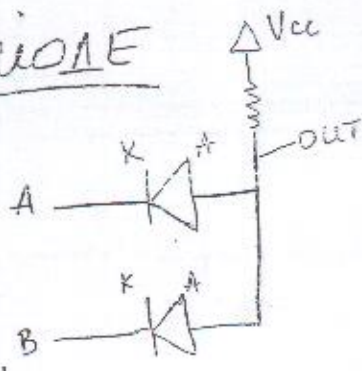
$$D_2: U_A = V_{CC}$$

$$U_K = 0$$

$U_{AK} = V_{CC} > V_{TH}$ проводи



АИОЛЕ



A	B	OUT
0	0	0
0	1	1
1	0	1
1	1	1

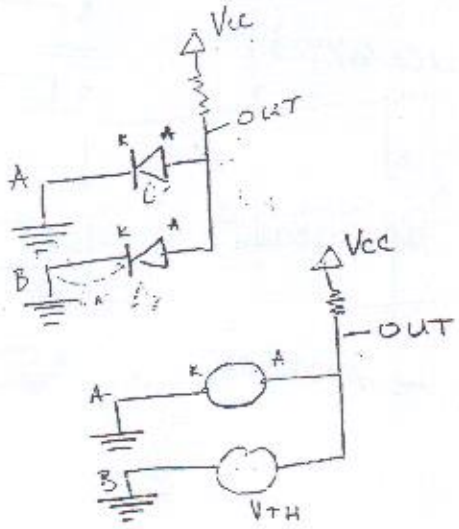
$$0 < V_{TH} < V_{CC}$$

TH - порог
Vcc

0 ПП: ОБЕ ИСКЛЮЧЕНЫ (УЗЕНЫ ИХ)

$D_1: U_A = V_{CC}$
 $U_K = 0$
 $U_{AK} = U_A - U_K = V_{CC} > V_{TH}$
 ПРОВОДИ

$D_2: U_A = V_{CC}$
 $U_K = 0$
 $U_{AK} = U_A - U_K = V_{CC} > V_{TH}$
 ПРОВОДИ

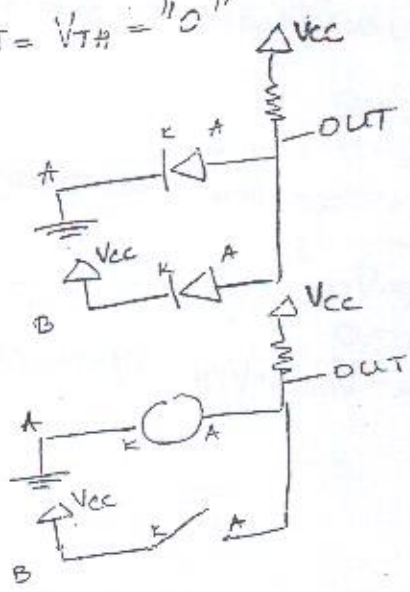


$OUT = V_{TH} = "0"$

1 ПП: D_1 ИСКЛЮЧЕНА, D_2 УКЛЮЧЕНА

$D_1: U_A = V_{CC}$
 $U_K = 0$
 $U_{AK} = V_{CC} > V_{TH}$
 ПРОВОДИ

$D_2: U_A = V_{CC}$
 $U_K = V_{CC}$
 $U_{AK} = 0 < V_{TH}$
 НЕ ПРОВОДИ

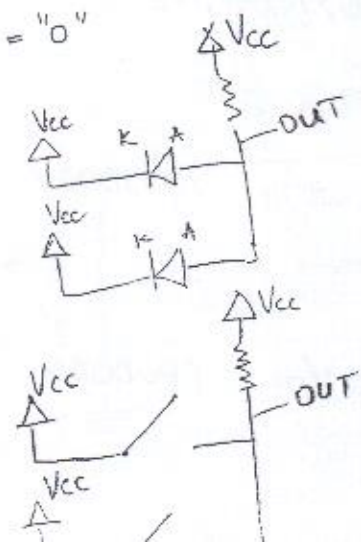


$OUT = "0V"$

1 = 10

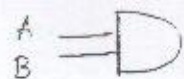
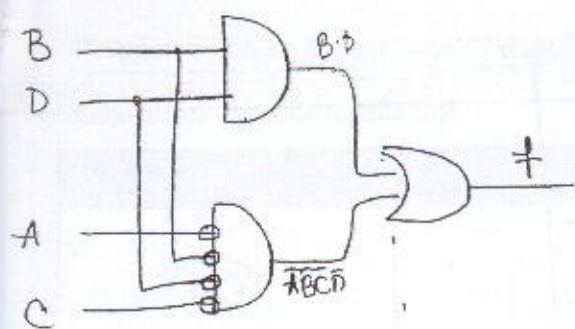
1 ПП: ОБЕ ДИОДЕ УКЛЮЧЕНЫ

$D_1: U_A = V_{CC}$
 $U_K = V_{CC}$
 $U_{AK} = 0 < V_{TH}$
 НЕ ПРОВОДИ



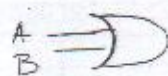
$OUT = V_{CC} = 1$

$$f = BD + ABCD$$



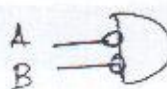
$$A \cdot B$$

"&"



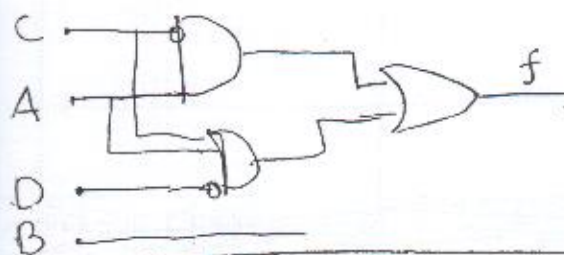
$$A + B$$

"|||"

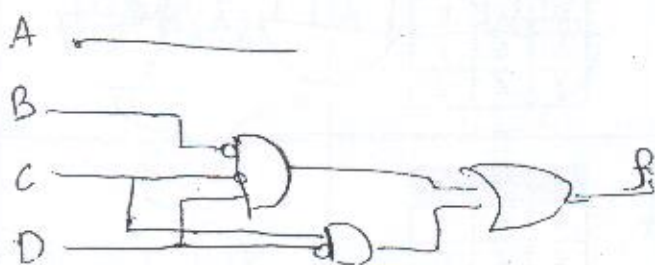


$$\overline{A} \overline{B}$$

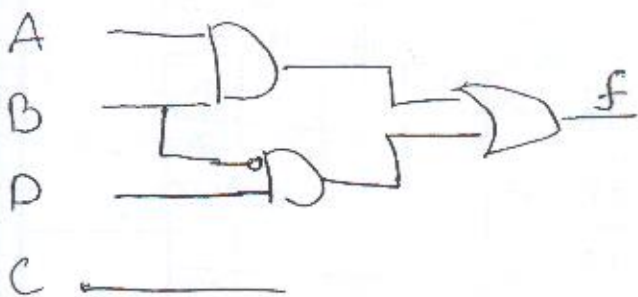
$$4. f = A\overline{C} + AC\overline{D}$$



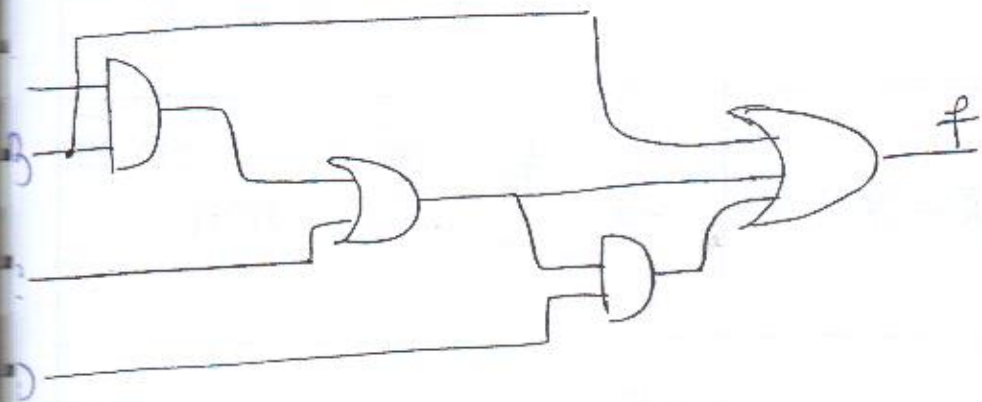
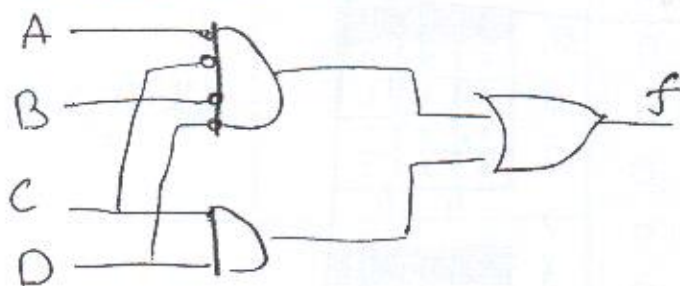
$$5. \overline{B}CD + C\overline{D}$$



$$f = AB + \overline{B}D$$



$$6. CD + \overline{A}\overline{B}\overline{C}\overline{D}$$

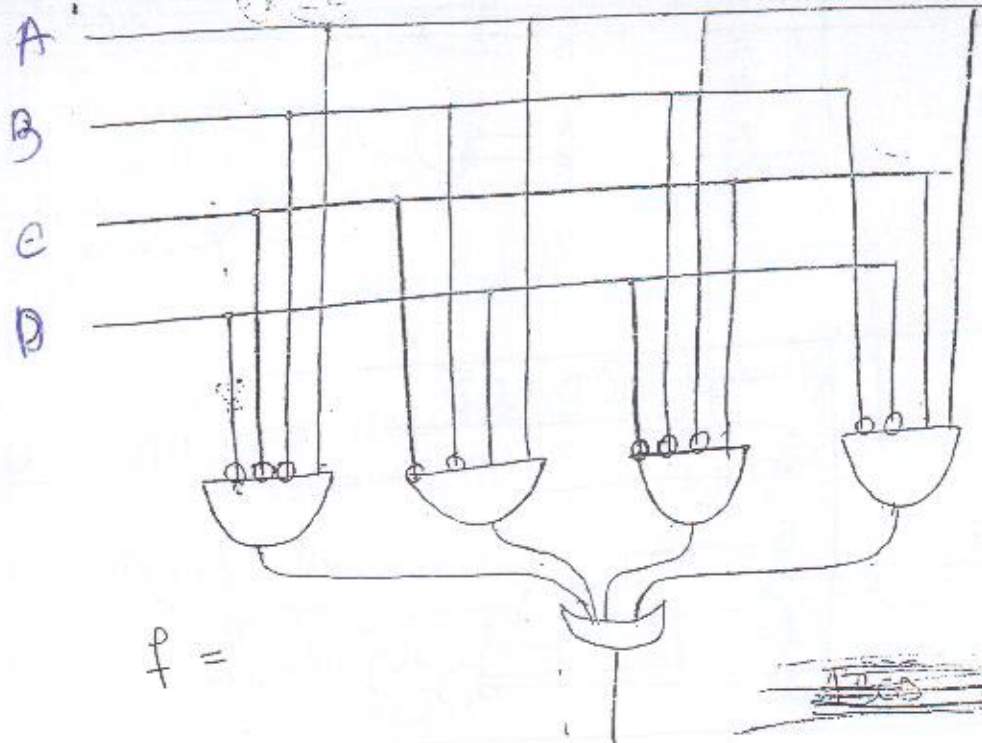


$$f = \overline{A}B + C$$

$$B$$

$$AB + C$$

$$(AB + C)D$$



A	B	C	D	y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

	CD	$\bar{C}\bar{D}$	$\bar{C}D$	$C\bar{D}$
AB	00	01	11	10
$\bar{A}\bar{B}$				*
$\bar{A}B$				*
$A\bar{B}$				
AB				
$\bar{A}\bar{B}$	*	*		*

$$f = A\bar{B}\bar{C} + C\bar{D}$$

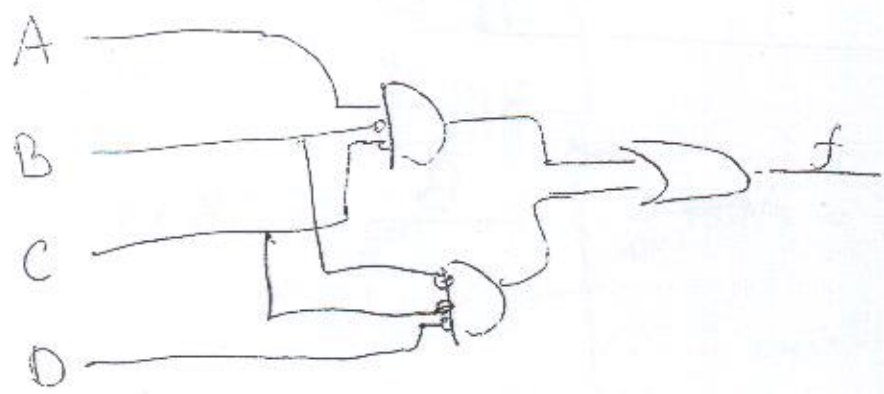


Tabela 1-1. Digitalna logička kola

Ime kola	Simbol	Funkcija	Istinitna tabela															
NOT		$Y = \bar{X}$	<table><tr><th>X</th><th>Y</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	X	Y	0	1	1	0									
X	Y																	
0	1																	
1	0																	
AND		$Y = X_1 X_2$	<table><tr><th>X₁</th><th>X₂</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	X ₁	X ₂	Y	0	0	0	0	1	0	1	0	0	1	1	1
X ₁	X ₂	Y																
0	0	0																
0	1	0																
1	0	0																
1	1	1																
OR		$Y = X_1 + X_2$	<table><tr><th>X₁</th><th>X₂</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	X ₁	X ₂	Y	0	0	0	0	1	1	1	0	1	1	1	1
X ₁	X ₂	Y																
0	0	0																
0	1	1																
1	0	1																
1	1	1																
XOR Cilob kombiniranih tache!		$Y = X_1 \oplus X_2$	<table><tr><th>X₁</th><th>X₂</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	X ₁	X ₂	Y	0	0	0	0	1	1	1	0	1	1	1	0
X ₁	X ₂	Y																
0	0	0																
0	1	1																
1	0	1																
1	1	0																
NAND		$Y = \overline{X_1 X_2}$	<table><tr><th>X₁</th><th>X₂</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	X ₁	X ₂	Y	0	0	1	0	1	1	1	0	1	1	1	0
X ₁	X ₂	Y																
0	0	1																
0	1	1																
1	0	1																
1	1	0																
NOR		$Y = \overline{X_1 + X_2}$	<table><tr><th>X₁</th><th>X₂</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	X ₁	X ₂	Y	0	0	1	0	1	0	1	0	0	1	1	0
X ₁	X ₂	Y																
0	0	1																
0	1	0																
1	0	0																
1	1	0																
XNOR		$Y = \overline{X_1 \oplus X_2}$	<table><tr><th>X₁</th><th>X₂</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	X ₁	X ₂	Y	0	0	1	0	1	0	1	0	0	1	1	1
X ₁	X ₂	Y																
0	0	1																
0	1	0																
1	0	0																
1	1	1																

SABIRANJE BINARNIH BROJEVA

$$\begin{array}{r} 1^o) \quad 1010 \quad (10) \\ + 0100 \quad (4) \\ \hline 1110 \quad (14) \end{array}$$

$$\begin{array}{r} 2^o) \quad 1001 \quad (9) \\ + 0011 \quad (3) \\ \hline 1100 \end{array}$$

$$\begin{array}{r} 3^o) \quad 0111 \quad (7) \\ + 0001 \quad (1) \\ \hline 1000 \end{array}$$

PROBLEM NASTAJE KADA:

$$\begin{array}{r} 1110 \\ + 0010 \\ \hline 10000 \end{array}$$

Rešenje: BROJ PRELAZI OPSEG
MORA SE PROŠIRITI

FRAKCIONI BITOVI (POSLE TACKE) $2^{-1} \quad 2^{-2} \quad 2^{-3} \quad 2^{-4} \quad 2^{-5}$

0,5 0,25 0,125 0,0625
↓
0,03125

$$0010 \cdot \underbrace{100}_{\uparrow} \Rightarrow 2.5$$

$$0101 \cdot \underbrace{110} \Rightarrow 5.75$$

$$0001 \cdot 001 \Rightarrow 1.125$$

$$1000 \cdot 101 \Rightarrow 8.625$$

011
100 + 1
101
proširiti se
mora

Pravilo:

$$\begin{array}{r} 1101, 011 \\ \hline S \end{array}$$

$$\frac{2^{-1}}{0,5} \quad \frac{2^{-2}}{0,25} \quad \frac{2^{-3}}{0,125} \quad \frac{2^{-4}}{0,0625} \quad \frac{2^{-5}}{0,03125} \quad \dots$$

piši koji su ovo brojevi: ako nije 6 onda 2 pred
crta 11625 - 1 = 112.625

S	1	1	1	1	F	F	F	Realan broj (Decimalno)
---	---	---	---	---	---	---	---	-------------------------

* Šta je ovo S?

$$f^1(ABCD) = (8, 9, 10, 12, 13, 14)$$

	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	$C\bar{D}$ 11	CD 10
00				
01				
11	*	*		*
10	*	*		*

0	1	3	2
4	5	7	6
12	13	15	14
8	9	11	10

$$f = A\bar{C} + C\bar{D}A$$

$$f^1(ABCD) = (1, 1, 1, 1, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0)$$

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

$$(15, 14, 13, 12, 11, 9, 3, 1)$$

	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	$C\bar{D}$ 11	CD 10
00		*	*	
01				
11	*	*	*	*
10		*	*	

0	1	3	2
4	5	7	6
12	13	15	14
8	9	11	10

$$f = AB + D\bar{B}$$

$$f^1(ABCD) = (0, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0)$$

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	$C\bar{D}$ 11	CD 10
00		*		*
01				*
11				*
10	*			*

0	1	3	2
4	5	7	6
12	13	15	14
8	9	11	10

$$f = C\bar{D} + \bar{C}D\bar{B}$$

$$f^1(ABCD) = (0, 3, 7, 11, 15)$$

	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	$C\bar{D}$ 11	CD 10
00	*		*	
01			*	
11			*	*
10			*	*

0	1	3	2
4	5	7	6
12	13	15	14
8	9	11	10

$$\bar{A}\bar{B}\bar{C}\bar{D} + CD$$

$$f^1(ABCD) = (1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 1, 0, 1)$$

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

$$(15, 13, 10, 8, 7, 5, 2, 0)$$

	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	$C\bar{D}$ 11	CD 10
00	*		*	
01		*	*	
11		*	*	*
10	*			*

0	1	3	2
4	5	7	6
12	13	15	14
8	9	11	10

$$f = \bar{A}\bar{B} + DB + \bar{C}\bar{D}\bar{B}\bar{D}$$

1 БУТ:

0	→	0
1	→	1

2 БУТА:

00	→	0
01	→	1
10	→	2
11	→	3

3 БУТА:

000	→	0
001	→	1
010	→	2
011	→	3
100	→	4
101	→	5
110	→	6
111	→	7

4 БУТА:

0000	→	0
0001	→	1
0010	→	2
0011	→	3
0100	→	4
0101	→	5
0110	→	6
0111	→	7
1000	→	8
1001	→	9
1010	→	10
1011	→	11
1100	→	12
1101	→	13
1110	→	14
1111	→	15

A
B
C
D
E
F

псте правило:

128	64	32	16	8	4	2	1
↑	↑	↑	↑	↑	↑	↑	↑
2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

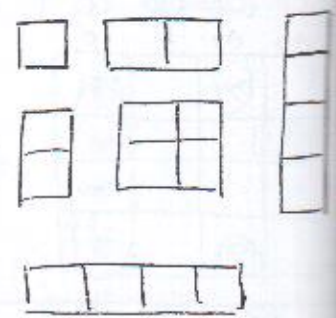
$f^1(ABCD) = (0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0)$

$f^1(ABCD) = (12, 11, 10, 4, 3, 2)$

	CD	C $\bar{D}$	C $\bar{D}$	CD	C $\bar{D}$
AB	00	01	11	10	
$\bar{A}\bar{B}$	00		*	*	
$\bar{A}B$	01	*			
AB	11	*			
$A\bar{B}$	10		*	*	

$f = B\bar{C}\bar{D} + \bar{B}C$

0	1	3	2
4	5	7	6
12	13	15	14
8	9	11	10



$\bar{A} \rightarrow 0$
 $A \rightarrow 1$

$f^1(ABCD) = (0, 5, 7, 13, 15)$

	CD	C $\bar{D}$	C $\bar{D}$	CD	C $\bar{D}$
AB	00	01	11	10	
$\bar{A}\bar{B}$	00	*			
$\bar{A}B$	01	*	*		
AB	11	*	*		

$f = BD + \bar{A}\bar{B}\bar{C}$

0	1	3	2
4	5	7	6
12	13	15	14
8	9	11	10